

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_4e32c8b8-725\agent-af94624f9a31c412c.jsonl`  
- SHA-256: `606238c3cab70bccd962424a31dc16f1b2722243506cbff0a81e53f57357cd99`  
- Source modified: `2026-06-19T17:20:17+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_4e32c8b8-725`  
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`
```

Transcript

1. user (2026-06-19T17:18:10.915Z)

You are auditing the docs/ folder of the badminton-highlight-indexer repo to decide which docs can move to docs/archives/. Be rigorous and EVIDENCE-BASED ? the owner has run two doc-staleness sweeps and cares deeply about not archiving live work or breaking links.

REPO ARCHIVE POLICY (docs/archives/README.md) ? APPLY STRICTLY:

- Archive ONLY work in a clear TERMINAL state: DONE / COMPLETED / REJECTED / fully SUPERSEDED. "Never archive in-flight or pending work."
- KEEP-LIVE (do NOT archive) anything that is: living/active, in-progress, a not-yet-started but still-intended design/plan (owner-gated future work is LIVE, not terminal), a reference/navigation doc (CODE_MAP, README, HOW_RALLY_DETECTION_WORKS, EVALUATION), a template (PROJECT_DOC_TEMPLATE), or a roadmap/backlog/strategy doc still being tracked.
- Destinations inside archives/: past_projects/ (bounded completed project / tech-debt / starter effort) · research/ (completed or superseded research notes/plans) · decisions/ (ADRs) · checkpoints/ (dated state snapshots) · archives/ top-level (one-off audits/reports/launch records). Naming convention for a completed doc: <NAME>-COMPLETED-<date>.md.
- ALREADY-ARCHIVED COPIES EXIST for these (so the top-level file is likely a pointer-stub or a stale DUPLICATE to reduce, NOT a fresh move): docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md, docs/archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md, docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md. If your doc already has an archived copy, check whether the top-level is a stub (leave it) or a full duplicate (flag DEDUPE).
- CONTRADICTION TRAP: a doc may say "COMPLETED" yet still be cited as a LIVE source elsewhere (e.g. CLAUDE.md cites DEFERRED_HARDENING_PLAN.md #16 as the next platform slice). Before recommending ARCHIVE, grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and other top-level docs for the filename. If it is referenced as a live/next source, verdict = ASK_OWNER (note the contradiction), not ARCHIVE_NOW.

YOUR BATCH (read each from docs/<name>): CODE_AUDIT_AND_TEST_HARDENING.md, DEFERRED_HARDENING_PLAN.md, HARDENING_LOOP_HANDOFF.md, EXECUTION_POLICY.md, REAL_FOOTAGE_VALIDATION_REPORT.md, R1_MILESTONE_LAUNCH_REPORT.md

For EACH doc:

1. Read the status header + TL;DR/intro + (if present) the \$7/\$8 progress tracker + definition-of-done. You do NOT need to read every line of long docs ? sample enough to judge lifecycle state honestly.
2. Decide lifecycle and verdict per the strict policy. Remember: owner-gated / not-yet-started DESIGN docs are LIVE future work (KEEP_LIVE), NOT terminal. Only DONE/COMPLETED/REJECTED/fully-superseded is terminal.
3. Check for an already-archived copy (the policy lists three; also Glob docs/archives for the doc's stem). If the top-level is a thin pointer-stub ? KEEP_LIVE (it's intentional). If it's a FULL DUPLICATE of an archived COMPLETED copy ? DEDUPE_STUB (recommend reducing top-level to a stub or removing).
4. Cross-check "cited as live": Grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and the other top-level docs/ for this filename. If it's cited as a live/next/authoritative source, it

CANNOT be archived even if it says COMPLETED ? ASK_OWNER with the contradiction noted.

5. Inbound references: Grep the whole repo (docs/, backend/, CLAUDE.md, *.py) for the filename to count link-breakage risk; list up to 6 key referrers.

Return the structured object with one entry per doc in your batch. Be decisive: most reference/template/active/design docs are KEEP_LIVE; reserve ARCHIVE_NOW for genuinely terminal single-efforts/reports/completed projects, and ASK_OWNER for the contradictory/ambiguous ones.

2. assistant (2026-06-19T17:18:13.198Z)

I'll audit each doc in my batch. Let me start by reading the archive policy, the key cross-reference files, and the docs themselves.

3. user (2026-06-19T17:18:16.611Z)

Your configuration specifies to merge with the ref 'refs/heads/docs/court-floor-rally-end' from the remote, but no such ref was fetched.

4. user (2026-06-19T17:18:16.855Z)

```
docs\archives\COMMERCIAL_READINESS_REVIEW.md
docs\archives\MONETIZATION_AUDIT.md
docs\archives\checkpoints\2026-06-eval-baseline\README.md
docs\archives\checkpoints\2026-06-logging-reporting\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md
docs\archives\past_projects\README.md
docs\archives\research\AUDIO_E1_FINDINGS.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
docs\archives\research\RALLY_DETECTION_RESEARCH.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\archives\quality-iterations\2026-06-initial\README.md
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\archives\decisions\DECISIONS.md
docs\archives\TEST_RUN_SUMMARY.md
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md
docs\archives\checkpoints\2026-06-14-hardening-loop-complete.md
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
docs\archives\HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md
docs\archives\README.md
docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\README.md
docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\lovo_resweep.py
docs\archives\quality-iterations\README.md
```

5. assistant (2026-06-19T17:18:20.558Z)

The pull failed (remote ref mismatch on the feature branch) ? proceeding read-only on local state, which is acceptable here. I note three already-archived copies exist for CODE_AUDIT_AND_TEST_HARDENING, DEFERRED_HARDENING_PLAN, and HARDENING_LOOP_HANDOFF. Let me read the policy and the docs in my batch.

6. user (2026-06-19T17:18:21.508Z)

```
1      # archives/
2
3      Umbrella directory for all **historical and completed** material.
4
5      ## Live vs Historical
6      - **Live docs** (for ramp-up, constant updates, and current work) live at the top level
of `docs/`.
7      - Everything here is no longer the primary source of truth. It is kept for context,
```

```

decision history, and regression detection.
8
9     ## Classification inside archives/
10    - `past_projects/` ? Completed small tech debt, starter projects, or low-regret
initiatives (e.g. the original low-regret moves + scaffolding work). These are "past" once they
reach a terminal state.
11    - `decisions/` ? Historical ADRs and major architectural decisions.
12    - `research/` ? Completed or superseded research notes and plans.
13    - `checkpoints/` ? Snapshots of major project states (especially useful for detecting
regressions in quality, priorities, or baselines over time as we do quality iterations).
14    - Other one-off audits, etc.
15
16    ## Archival Rule
17    Only move something into `archives/` (or its subdirectories) when the work is in a clear
**terminal state**:
18    - DONE
19    - COMPLETED
20    - REJECTED
21
22    Never archive in-flight or pending work.
23
24    When archiving a project/phase:
25    1. Ensure the main document has an explicit `**STATUS: COMPLETED**` (or DONE/REJECTED)
header at the top.
26    2. Link back to the current living documents in `../` (e.g. `COMMERCIALIZATION.md`,
`BACKLOG.md`).
27    3. For regression detection value, consider adding a dated entry under `checkpoints/`.
28
29    See `past_projects/README.md` for the specific guidance on that category.
30

```

7. user (2026-06-19T17:18:21.700Z)

```

1     # Code Audit, Test Infrastructure, Positive Testing Plan, and Regression Hardening
(COMPLETED)
2
3     **Status: COMPLETED** (2026-06-14; all T1?T5 positive tests + verification + loop
complete per #162/#163; archived snapshot in `docs/archives/`)
4
5     **Owner:** avidullu (with agent assistance)
6
7     **Related:** See the full archived snapshot at
`docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` and
`docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md`.
8
9     ---
10
11    **Note (per review on finale PR #165):** This top-level file is now a short pointer stub
to avoid content duplication. The complete living document (Executive Summary, all Audit
Findings with file:line, Task List, Detailed Positive Test Plans §4, Real-Video Protocol §5,
Ablation integration §6, Verification Guard §7, and the full Living Log §8 with every
pre/post/review/post-merge record) lives *only* in the archived snapshot.
12
13    The initiative is complete. See the archives/ copy and checkpoint for the final state
and verification details.
14

```

8. assistant (2026-06-19T17:18:24.234Z)

CODE_AUDIT_AND_TEST_HARDENING.md is already a thin pointer stub (intentional, per PR #165).
KEEP_LIVE. Now the other docs.

9. user (2026-06-19T17:18:25.185Z)

```

1      # Deferred Hardening Plan ? Train/Eval Guardrails, Async Jobs, Packaging (COMPLETED)
2
3      **STATUS: COMPLETED** (2026-06-14). This planning doc's three items are all dispatched:
4
5      - **#16 Async job model ? DONE** (PR #87; hardened by the self-scheduling drain loop in
6      #149).
7      - **#7 Packaging modernization ? DONE** (PR #167: PEP 621 `pyproject.toml`, `setup.py` ?
8      `scripts/doctor.py`, `indexer` console entry point, CI switched to `pip install -e .[dev]`).
9      - **#13 Train/eval separation guardrails ? moved to [BACKLOG.md](BACKLOG.md)** (?; the
10     concrete `splits.py` / `assert_disjoint` match-level-split design is preserved there).
11
12     Full archived snapshot (the original plan + per-item design/risks/verification):
13     [archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md](archives/DEFERRED_HARDENING_PLAN-
14     COMPLETED-2026-06-14.md).
15
16     _(This top-level file is a pointer stub kept so existing links to
17     `docs/DEFERRED_HARDENING_PLAN.md` still resolve; the content now lives in the archive.)_
18
19

```

10. user (2026-06-19T17:18:25.811Z)

```

1      # Hardening Loop Handoff & Restart Prompt (COMPLETED)
2
3      **This is the contract for the PR babysit loop (T1-T5 + finale).**
4
5      Full snapshot of the final version (including all state transitions, the correction for
6      #165 babysit, and the exact bootstrap prompt) is archived at
7      `docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md`.
8
9      ---
10
11     **Note (per review on finale PR #165):** Top-level reduced to pointer stub to have
12     single source of truth in archives/ (no duplication with the archived copy). The scheduler
13     bootstrap text, strict rules, and current STATE (finale PR #165 babysit until merge) are
14     preserved in the archived snapshot.
15
16     **All items done.** #165 (finale PR) merged per owner LGTM (comment 4702233617).
17     Initiative archived, clean slate. No further schedulers. Per handoff rules and #162.
18
19     **#166 merged** (c558372). Transition record + P3 signal + work-items plan (with risk
20     assessment, milestones, sequencing for review-first execution) in
21     `docs/POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md`. Loop stopped per owner directive
22     (no further schedulers). See that plan MD for full details and "How to use this plan".
23
24     (The full rules and historical STATE snapshot remain in the archived copy per the #165
25     review intent to keep top-level a short pointer.)
26
27

```

11. assistant (2026-06-19T17:18:29.143Z)

Both DEFERRED_HARDENING_PLAN.md and HARDENING_LOOP_HANDOFF.md are already thin pointer stubs. But the policy flags a CONTRADICTION TRAP: CLAUDE.md cites DEFERRED_HARDENING_PLAN.md #16 as the next platform slice. Let me check that and read the remaining three docs.

12. user (2026-06-19T17:18:30.105Z)

```

1      # Execution policy ? per-job, self-scheduling resilience for long/flaky operations
2
3      **Status:** **P1 SHIPPED** (the actual hang fix); P2 + P3a substrate landed; P3b next.
4      Motivated by a
5      real GX010125 hang: the Gemini `generateContent` call stalled with **no client-side
6      timeout** and
7      blocked the whole run. The fix is not a one-off timeout ? it is a **policy** every long
8      operation
9

```

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6      carries, so waiting / polling / aborting / resuming is declared per job, not hard-coded.
P1 lives in
7      `backend/infrastructure/execution_policy.py` (`ExecutionPolicy` + `run_bounded` /
`poll_until`); the two
8      unbounded Gemini waits are now bounded in `backend/providers/gemini.py` (the file-
processing poll under
9      `poll_until`, `generate_content` under `run_bounded` with `op_timeout_sec`). Remaining
slices (P3b?P4)
10     stay owner-gated.
11
12     ## The problem (concrete)
13     `backend/providers/gemini.py::analyze_video` has two unbounded waits:
14     1. the file-processing poll ? `while video_file.state.name == "PROCESSING":
time.sleep(10)` (no deadline);
15     2. the `generate_content` call ? the retry loop (6 attempts, backoff) catches
failures, not hangs;
16     a request that never returns blocks forever.
17     The same shape recurs anywhere we wait on an external/long thing (WSL/WASB GPU pass,
ffmpeg, a future
18     remote compute backend). Today each site invents its own ad-hoc loop (or none). We want
one declarative
19     contract.
20
21     ## Your architecture ? validated
22     - "Each job can specify its own [policy]." ? Yes ? an `ExecutionPolicy` is attached
per job
23     (and per long sub-operation), defaulting sensibly, overridable by the caller. It is
just data on the job.
24     - "Jobs can self-schedule." ? Yes ? a job that must wait persists its `next_poll_at`
+ progress and
25     re-enqueues itself; the existing single worker re-claims it when due. No periodic
central tick.
26     - "A master node is not necessary." ? Correct. The repo is already master-less: a
`ThreadPoolExecutor`
27     worker (`main.py`, `max_workers=1`, single self-hosted box) drains the `jobs` table,
and a crash-recovery
28     sweep (`fail_stale_jobs()`, DEFERRED_HARDENING #16) reaps jobs a dead process left
behind. Self-scheduling
29     = cooperative re-enqueue; any worker can claim a due job, so it generalizes to N
workers
30     (PLATFORM_ARCHITECTURE pluggable compute) without a dedicated scheduler ? the DB
is the coordination.
31
32     ## The contract
33     ```python
34     class WaitMode(str, Enum):
35         WAIT_AND_RESUME = "wait_and_resume" # DEFAULT ? bounded wait; on deadline, persist
+ reschedule
36         ABORT = "abort" # on deadline/hang, stop this op (fail or
partial)
37         BLOCK = "block" # legacy in-process bounded blocking wait (no
reschedule)
38
39     @dataclass(frozen=True)
40     class ExecutionPolicy:
41         mode: WaitMode = WaitMode.WAIT_AND_RESUME
42         poll_every_n_sec: float = 10.0 # cadence for polling an in-progress external
op
43         op_timeout_sec: float = 120.0 # HARD per-attempt deadline ? the hang-killer
(was missing)
44         max_wait_sec: float = 1800.0 # total wall budget across resumes before
giving up
45         max_retries: int = 5 # failures (not hangs); exponential backoff +
jitter
46         backoff_base_sec: float = 3.0

```

```

47         on_timeout: Literal["reschedule", "partial", "fail"] = "reschedule"
48         resumable: bool = True # persist partial progress so a resume
continues, not restarts
49         # JSON-serializable ? rides on the job row; a variant/job overrides any field.
50         ...
51         - **`op_timeout_sec`** is the missing piece that would have killed the GX010125 hang:
every external call
52         (upload, `files.get`, `generate_content`) runs under a hard per-attempt deadline; on
expiry the policy's
53         `on_timeout` decides ? `reschedule` (WAIT_AND_RESUME), `partial` (proceed with what's
done), or `fail`.
54         - **Hang vs failure are distinct.** Retries (`max_retries` + backoff) handle a call that
*errors*; the
55         *timeout* handles a call that *hangs*. Both are policy, not provider-baked.
56
57         ## Self-scheduling mechanism (no master)
58         A long op under `WAIT_AND_RESUME`:
59         1. runs the external call under `op_timeout_sec`;
60         2. on success ? record progress, continue;
61         3. on timeout/in-progress ? **persist progress + `next_poll_at = now +
poll_every_n_sec`, set job
62         `status=waiting`, release the worker** (don't hold a thread sleeping for minutes);
63         4. the worker loop (and the startup sweep) claims jobs whose `next_poll_at <= now` and
resumes them;
64         5. give up when cumulative wait > `max_wait_sec` ? `on_timeout` terminal action.
65
66         This needs **only** two `jobs`-table columns (`next_poll_at`, `progress` JSON ?
`progress` already exists)
67         and a "claim due jobs" query. No timer service, no leader, no master. On one box it is
one worker draining a
68         priority-by-due-time queue; on many boxes it is the same query with a row-claim (`UPDATE
? WHERE status=
69         'waiting' AND next_poll_at<=now`), so a job migrates to whoever is free.
70
71         ## Resumability (why WAIT_AND_RESUME is the default)
72         The op persists **what it already finished** so a resume does *not* redo it. For the AI
handoff that means
73         **per-window confirmation state**: confirmed windows are written as they complete; a
resume processes only
74         the unconfirmed ones. The GX010125 run is the worked example ? instead of the 7-min cap
abandoning ~10
75         windows and losing them, WAIT_AND_RESUME would persist the confirmed windows and
reschedule the rest, so the
76         reel becomes exhaustive across resumes with **no wasted Gemini spend**. `ABORT` (or
`on_timeout=partial`) is
77         the opt-out for "good enough now" (what the one-off script does today).
78
79         ## Phasing (owner-gated; ONE PR per slice, off `master`, default = current behaviour)
80         - **P0 ? this doc.**
81         - **P1 ? `ExecutionPolicy` + bound the two Gemini waits (the actual hang fix) ?
SHIPPED.** The dataclass
82         lives in `backend/infrastructure/execution_policy.py` (`run_bounded` = HARD per-
attempt
83         `op_timeout_sec` + retry/backoff; `poll_until` = bounded readiness poll).
`GeminiProvider.analyze_video`
84         now runs the `files.get` processing poll under `poll_until` and `generate_content`
under `run_bounded`
85         (`op_timeout_sec=180`), so a hung call terminates per `on_timeout` instead of blocking
forever ? while
86         the default policy otherwise reproduces today's behaviour. Pure-logic, unit-testable
(a fake provider
87         that hangs ? policy aborts/reschedules).
88         - **P2 ? per-job policy substrate (DB) ? SHIPPED.** `jobs` table `policy` +
`next_poll_at` columns
89         (user_version 3), `claim_due_job` (atomic, earliest-due, compare-and-set),

```

```

`set_job_waiting`,
90     `set_job_policy`. These are the primitives; nothing drove them yet.
91     - **P3a ? worker self-scheduling LOOP ? SHIPPED.** The `main.py` worker now DRAINS the
table via
92     `claim_due_job` (`_drain_due_jobs`) instead of the one-shot `submit(_run_job, ?)`, and
honours
93     WAIT_AND_RESUME: a job raising `JobWaiting(delay_sec, progress)` is parked
(`set_job_waiting` ? status
94     `waiting` + `next_poll_at`, worker released) and re-claimed when due. A daemon wake-
timer
95     (`_arm_wake_timer`, delay from `seconds_until_next_due`, capped) and a startup drain
resume parked jobs
96     with no client submit ? no central scheduler (the DB is the clock). Additive: the
production segmenter
97     path never raises `JobWaiting`, so today's jobs run unchanged.
98     - **P3b ? resumable AI handoff** (persist per-window confirmation; resume processes only
unconfirmed; a
99     long op exhausting its in-process wait budget raises `JobWaiting` from inside the
segmenter) ? makes the
100     GX010125-style run exhaustive across resumes. **NEXT slice.** Note: a `JobWaiting`
raised from inside the
101     segmenter currently rides the destroy-after-confirm prune; P3b must persist confirmed
windows ABOVE that
102     watermark (or skip the prune on park) so a resume does not redo confirmed work.
103     - **P4 ? apply the same policy to the WSL/WASB GPU pass + ffmpeg** (the other long ops).
104
105     ## Trade-offs / notes
106     - **Default WAIT_AND_RESUME, but bounded** ? `max_wait_sec` guarantees forward progress
/ eventual give-up;
107     it never waits truly forever (the current bug).
108     - **Idempotent resumes** ? a resumed op must not double-spend (e.g. re-upload a clip
already confirmed);
109     P3's per-window state is what makes that safe.
110     - **Cost control** ? `op_timeout_sec` + `partial` is the cheap path; `WAIT_AND_RESUME`
trades wall-clock for
111     completeness without re-spending. Both are one field on the job.
112     - **No new infra** ? rides the existing `jobs` table + single-worker executor + crash
sweep; the DB is the
113     coordinator, so it scales to multi-worker (PLATFORM_ARCHITECTURE) with no master
added.
114
115     **Links:** the async-job substrate is `docs/DEFERRED_HARDENING_PLAN.md` §3 (#16); multi-
worker future is
116     `docs/PLATFORM_ARCHITECTURE.md`; the motivating hang is the GX010125 run
117     (`scratch/run_gx010125_full.py`, where a 7-min cap had to be hand-rolled).
118

```

13. user (2026-06-19T17:18:30.576Z)

```

1     # Local (No-AI) Rally Detector ? Real-Footage Validation & Improvement Roadmap
2
3     **Date:** 2026-06-16 . **Audience:** owner/cofounder . **Status:** validation complete,
roadmap proposed (all flags default-OFF)
4     **Footage:** 6 Kushagra/Yash June-12 clips, 1080p proxies, fully-local no-AI WASB
windowing vs Gemini (strong reference, **not** ground truth except mahadevpura-2, owner hand-
labeled).
5     **Tracked doc to fold this into:** `docs/GOLDEN_REGRESSION_FIXTURES.md` . root-cause
backbone: `docs/RALLY_QUALITY_RESEARCH.md` §2.
6
7     ---
8
9     ## 1. Headline verdict
10
11     The W1 `max_window_duration` cap **fixed the over-merge catastrophe** ? aggregate count
went from 48 (29% of Gemini's 165) to 250 (152%), and mIoU rose 0.23 ? 0.39 ? but it **overshot

```

into over-segmentation and never landed near 1.0**, so we traded a merge problem for a split problem rather than solving localization. On the one clip with owner ground truth (mahadevpura-2), the best local config is **~86% of Gemini at loose tolerance (F1@0.25 0.722 vs 0.835) but only ~53% at tight boundaries (F1@0.5 0.296 vs 0.557)** ? and the entire residual gap is **rally-END over-extension** (|?end| 3.12s vs Gemini 1.19s), while rally-START is already near Gemini parity (|?start| 1.74s vs 1.46s).

12
13 ---
14
15 ## 2. What worked ? the over-merge fix at scale
16

17 The baseline collapsed entire games into mega-windows. The cap demolished that pathology:

18
19 | Metric (aggregate, n=6) | Baseline | W1 (cap=12) |
20 |---|---|---|
21 | Rally count | 48 (0.29× Gemini) | 250 (1.52×) |
22 | Mean window length | 62.9s | 10.8s |
23 | Merge-groups | 27 | 10 |
24 | mIoU | 0.23 | 0.39 |

25
26 - The signature pathology ? **GX010129: 2 windows, mean 267s, 27 Gemini rallies trapped in one merge, mIoU 0.0** ? is gone (W1: 50 windows).

27 - On the ground-truthed clip, baseline F1@0.5 was **0.000** (mIoU 0.03); the best local config reaches **0.296** (mIoU 0.50). That is the real, measured win: from "unusable" to "detects most rallies, boundaries still loose."

28 - **Detection recall is genuinely good:** at ±0.25 tolerance, local finds ~86% of what Gemini finds on the ground-truthed clip. The remaining problem is *where the boundaries land*, not *whether rallies are found*.

29
30 ---
31
32 ## 3. What broke / new problems
33

34 **(a) The over-segmentation pendulum is real and global.** No single global setting hit ~1.0×: baseline 0.29× ? W1 1.52× ? W1+W2 1.35×. The cap swung us straight past parity into over-splitting.

35
36 **(b) Per-venue variation defeats any single global config.** Under *identical* W1 knobs, per-clip count ratio ranges **0.22× ? 2.00×** and mean window length **7.8s ? 86.9s**:

37
38 | Clip | Gemini | W1 | ratio | mean window |
39 |---|---|---|---|---|
40 | GX010128 | 39 | 78 | 2.00× | 7.8s |
41 | GX010129 | 27 | 50 | 1.85× | 9.3s |
42 | GX020125 | 11 | 18 | 1.64× | 14.1s |
43 | mahadevpura-0 | 40 | 62 | 1.55× | 10.4s |
44 | mahadevpura-1 | 9 | 2 | 0.22× | **86.9s** (blob) |
45 | mahadevpura-2 | 39 | 40 | **1.03×** | 13.8s |

46
47 Tightening to fix GX over-split pushes mahadevpura-0/-1 further under; loosening to merge the blob worsens GX. The root cause is mechanical: the active-frame rule keys on **"shuttle moving," not "rally"**. ? a window ends only when the shuttle goes invisible or drops below `velocity\_thresh` (`tracknet\_runner.py:305`, end = last in-flight frame; `:285`, `prev` resets on any invisible frame; `:292`, active requires velocity > 0.02). So inter-clip differences in *how the shuttle behaves between points* ? not rally structure ? drive the result. **Honest caveat:** this correlates with clip content/activity structure, **not cleanly with venue** ? mahadevpura-0 over-splits like the GX clips (1.55×), so a "GoPro vs court" venue story is *not* supported at n=3/venue.

48
49 **(c) The continuously-active blob (mahadevpura-1) is a distinct, unsolved failure mode.** The clip has no inactivity gap ? `merge\_gap` to split on, so ~174s coalesces. W1's cap *did* fire once (split into ~101.9s + ~71.9s) but both halves stay ~678× over the 12s cap ? there is no sub-gap to recurse on. Result: n=2, 0 matched, mIoU 0.0. The cap is **under-effective, not a no-op** here; W2 is a true no-op (W1+W2 byte-identical). This single clip's

mIoU 0.0 must not be averaged into the headline.

50

51 **(d) W2 (net-crossings gate) buys count by spending recall, and only fires on one venue.** W1 ? W1+W2 dropped aggregate count 250 ? 222 (?28), but **~all of it is mahadevpura-0** (62?43, ?19, with **0?5 new gemini-only misses**); GX clips moved ?3 each, the blob 0. mIoU stayed flat at 0.39 and **6 net-new gemini-only misses** appeared. Golden lift was only **+0.019 (n.s.)** vs W1's +0.139. It violates do-no-harm for no ground-truth-backed gain.

52

53 **(e) Count parity is a deceptive proxy.** mahadevpura-2 lands at 1.03x ? near-perfect count ? yet matches only **12/39** with mean_iou 0.420 and F1@0.5 0.150 (W1+W2). Right count, wrong boundaries.

54

55 **(f) tIoU@0.5 over-penalizes and flips rankings.** For a 5.7s rally, tIoU 0.5 tolerates only ~±1.9s, and the ~1?1.5s service window nearly exhausts that budget. The hard-cap is golden-neutral at n=6 (+0.006 n.s.) but becomes *best* at n=7 once the continuously-active clip enters ? a metric artifact, not a real flip in quality.

56

57 ---

58

59 ## 4. The named next lever ? a rally-END signal (not a generic boundary head)

60

61 **Lever: a sustained-inactivity rally-END rule (See et al. style) in `trajectory\_to\_action\_windows`.**

62

63 **Why this specifically, and why not a generic boundary refiner:** the ground-truth boundary errors decompose cleanly ?

64

65 - **rally-START is already solved:** local |?start| median **1.74s ? Gemini 1.46s**.

66 - **rally-END is the entire gap:** local |?end| median **3.12s ? Gemini 1.19s**.

67

68 A generic "tighten both boundaries" head spends half its effort on START, which is already at reference parity ? wasted ROI and risk of regressing a good edge. The asymmetry says: **end the window at the last *sustained-active* frame, not at the last in-flight frame carried to the cap.**

69

70 Concretely: maintain a trailing window T (~1.5?2s); terminate the active group once ?80% of its frames are no-motion (velocity ? `velocity\_thresh` OR shuttle invisible), setting end at the *start* of that inactivity run rather than `group[-1][0]/fps` (:305`). Critically, \*\*decouple the END decision from single-frame visibility\*\* ? stop treating one invisible frame as a hard trajectory break (:285`) for the inactivity counter; count short occlusion bursts as "no-motion" contributions so brief end-of-rally occlusion doesn't cut early and a slow walk-back doesn't extend the window. This should tighten END on *every* venue without touching count on the well-behaved clips.

71

72 **Honest caveat on the END diagnosis:** the start?end-parity decomposition rests on the n=40 mahadevpura-2 ground-truth numbers above. An earlier, more aggressive framing (a cited `output/local\_vs\_gemini\_6.json` with per-matched-pair start/end stats across all 6 clips) **does not exist in this worktree** and must not be cited as if it does. The single-clip GT we *do* have is consistent and points the same direction, but the END-dominance claim is currently **n=1 ground-truth** ? confirm on GX010128 before treating it as settled.

73

74 ---

75

76 ## 5. Prioritized roadmap

77

78 | # | Lever | Effort | Expected payoff | Confidence | Depends on |

79 |---|---|---|---|---|---|

80 | **R1** | **Ship the smallest safe milestone as ONE flip: W1 cap=12 ON + W1.5 hard-cap (midpoint-chop when no gap) ON, no W2.** Best-measured local config; hard-cap fires only when W1 finds no gap, so it's a no-op on the 4 gap-segmented clips. Assert never-empty (?1 window, `min\_window`=2.0s) as a tested invariant. | S | F1@0.5 0.150?0.296, F1@0.25 0.500?0.722, mIoU 0.31?0.50 on the GT clip; rescues the blob from the degenerate 1?2 window outcome | High (measured on GT clip) | ? |

81 | **R2** | **New primary eval metric: ±2s tolerance-match F1 + boundary-error-in-seconds, with over-seg guard.** Asymmetric Hungarian one-to-one match (?_start=2.0s,

?_end=1.0?1.5s); report P@?, R@?, F1@?; split/merge counts per video; signed+abs ?start/?end median+IQR. Demote tIoU/mIoU to secondary diagnostic. | S?M | Stops the count-parity trap and the tIoU over-penalty; makes over-seg cost precision directly; tells us start is solved / end is open | High | ? |

82 | **R3** | **Re-scope the cap to true rally length** ? test `max\_window` in the **6?8s** band (+ separate end-trim) instead of 12s; the hard-cap lift already signals 12s is coarse. Ablate cap-only vs hard-cap-chop on the GT clip, holding velocity_thresh/merge_gap fixed. | S?M | Less END over-extension and less over-counting on short-rally venues | Medium (12s coarseness measured; 6?8s untested) | R2 (need the metric to score it) |

83 | **R4** | **Sustained-inactivity END rule** (§4): trailing-window 80%-no-motion termination; decouple END from single-frame visibility. Config-gated, default-OFF. | M | Directly attacks the |?end| 3.12s?target ?~1.2s (Gemini parity); helps every venue, count-neutral on clean clips | Medium-High (root-cause-aligned; needs n>1 GT) | R2, R3 |

84 | **R5** | **Blob splitter for the gap-less case** (mahadevpura-1): when the inactivity rule never fires within an over-cap group, fall back to W1.5 midpoint-chop **and log it as a forced cut** flagging the clip as needing rally-STATE signals (net-crossings / two-sided motion), not a bigger cap. Track as its own eval stratum. | M | Removes the mIoU 0.0 / 0-match pathology on continuously-active clips; prevents one clip swinging the mean | Medium | R1, R4 |

85 | **R6** | **Expand + stratify golden set to ~16 by the weekend**, over-sampling continuously-active clips and each venue. Re-fit ? against a genuine inter-annotator study (2nd labeler re-labels 2?3 Roora videos; set ? to 90th-pct pairwise human \||?|). Re-run LOVO + sign-test before locking any default. | M | Stops single-clip rank flips (hard-cap n.s.@6 ? best@7); makes ? defensible | ? | R2 |

86 | **R7** | **Keep W2 (net-crossings) OPT-IN / default-OFF**, retained as a flag-gated scorer/post-filter so it keeps accruing measurements (weak-signals-retained directive). Do *not* promote. | S | None now; future signal for the corpus | High (the harm is measured: 6 net-new misses, +0.019 n.s.) | ? |

87 | **R8** | **Do NOT build a serve/START classifier** (See et al. M1) until END lands. START is at Gemini parity; marginal ROI is on END. Revisit only after END closes. | ? | Avoids wasted effort | High | R4 outcome |

88

89 **Default-ON config is GLOBAL, not per-venue.** The hard-cap's fire-only-when-no-gap behavior is already content-adaptive, so it covers the continuously-active case without a brittle per-venue switch fit on n=1. Revisit per-venue presets only when the corpus has meaningful per-venue n (well beyond ~16).

90

91 ---

92

93 ## 6. Eval methodology change (gate the roadmap on this)

94

95 Replace headline tIoU-F1 with a **task-faithful tolerance metric**:

96

97 1. **Tolerance-match F1 (PRIMARY).** A prediction matches a reference iff `|?start| ? ?\_start AND |?end| ? ?\_end`, asymmetric: **?_start = 2.0s** (~2× the service-window/inter-annotator start noise; Gemini ?start median 1.46s), **?_end = 1.0?1.5s** (rally-end is a sharp event; Gemini ?end median 1.19s, and it's exactly where local fails at 3.12s). Freeze these as a documented decision in `docs/GOLDEN\_REGRESSION\_FIXTURES.md` with the IoU?time derivation `? = D.(1??)/(1+?)` recorded so the choice is auditable, until R6's inter-annotator study refines them.

98 2. **Over-segmentation guard (MANDATORY).** Strict one-to-one matching via **Hungarian global assignment, never greedy.** Report **P@? = matched/n_pred** alongside **R@? = matched/n_ref**. This makes the 222/165 = 1.35 surplus cost precision directly (mahadevpura-2: 40 preds vs 40 refs, 12 matched ? P=0.30, R=0.30 ? over-seg fully visible; widening ? can't hide it).

99 3. **Fragmentation/merge pair (SECOND guard, no-regression-gated per video).** split_count (predictions covering one reference ? catches W1 over-seg) and merge_count (references under one prediction ? catches the baseline mega-window pathology). **A promotion may not increase either guard on ANY video even if mean F1 rises** ? this blocks the mahadevpura-0 trap (count parity improved 1.55?1.075 while gemini_only misses went 0?5).

100 4. **Boundary error as TWO separate distributions** (median + IQR of signed and absolute ?start, ?end over matched pairs), never folded into one IoU. This is the metric that says start is solved and end is the open problem, and directs effort to the velocity-decay / net-crossing tail.

101 5. **tIoU/mIoU stays as a SECONDARY diagnostic** ? single-number trend line, ties to the paper ablation backbone ? but is **not the gate.**

```

102
103     **Promotion (trustworthiness) rule at n=6??:** a delta is promotable only if *(1)** the
paired per-video sign-test on tolerance-F1 is a near-unanimous sweep ? **6/0 at n=6 (p=0.031)**
or **7/0 at n=7 (p=0.016)**; 5/1 or 6/1 is NOT significant (p=0.219 / 0.125); **AND (2)** over-
seg and merge guards do not regress on any video; **AND (3)** the effect size is non-negligible
(mean delta + bootstrap CI as an honesty band ? expect it to span 0 at this n; use it to
separate baselineW1 +0.139 "real" from W1?W1+W2 +0.019 "directionally-clean-but-negligible,"
not as a pass/fail gate). **Stratify, don't blend:** report mahadevpura-1 as its own stratum.
104
105     ---
106
107     ## 7. Honest caveats
108
109     - **n=6 clips, exactly ONE with ground truth (mahadevpura-2).** Every tight-boundary
number (F1@0.5, |?start|/?end|, the start-solved/end-open decomposition) rests on that single
owner-labeled clip. **GX010128 is the next ground-truth target** to confirm the END-dominance
diagnosis on a second, denser clip before locking R4.
110     - **Gemini is a strong REFERENCE, not ground truth.** The other 5 clips are scored
against Gemini; "1.52x of Gemini" measures agreement with Gemini, not correctness. Gemini stays
strictly an **inference-time reference, never a training-data source** (licensing guardrail).
111     - **Single-clip effects are load-bearing and fragile.** W2's count reduction is
~entirely mahadevpura-0; the hard-cap's "best at n=7" status is driven by the continuously-
active clip; the blob's mIoU 0.0 can swing a 6-clip mean. This is exactly why R6 (stratify +
expand to ~16) gates locking any default.
112     - **Do not over-claim "at par."** Local is **~86% of Gemini on rally DETECTION (F1@0.25
0.722 vs 0.835)** but only **~53% on tight boundaries (F1@0.5 0.296 vs 0.557)**. START is at
parity; the headline number people will quote should be the honest split, not a single rounded
figure.
113     - **Refuted claims explicitly NOT relied on:** there is no
`output/local_vs_gemini_6.json` in this worktree, so any per-matched-pair start/end stats
*across all 6 clips* are unsubstantiated; the only local DB present (`output/val.db`) covers
GX010128, GX010129, mahadevpura-0 (not mahadevpura-2). The "cap=12 is the proximate cause of END
over-extension" and "GX010128 chops single true rallies" claims are **not** supported by the
present data (GX010128's inflation is dominated by 34 local_only false-positive windows, splits
metric only 4; its W1 windows average 7.76s, *longer* than Gemini's 5.85s). Treat cap=12 as *one
contributor* alongside detector recall and the 2s merge_gap, to be isolated by R3's ablation ?
not as the established root cause.
114     - **All proposed flags ship default-OFF**, promoted only on the $6 rule. The only thing
recommended for default-ON is the R1 milestone (W1 + hard-cap, no W2), and even that is **gated
on a re-run at the ~16-clip corpus** since the hard-cap was golden-neutral at n=6 and only
became best at n=7.
115
116     ---
117
118     *Files referenced (absolute):*
119     `C:\Users\avidu\Projects\badminton-highlight-
indexer\claude\worktrees\w2-safeguard\docs\GOLDEN_REGRESSION_FIXTURES.md` (tracked doc to fold
this into) · `...\docs\RALLY_QUALITY_RESEARCH.md` $2 (root-cause backbone) · code anchors in
`backend\pipeline\segmenters\...\tracknet_runner.py` lines 285, 292, 305, 358?364.

```

14. user (2026-06-19T17:18:31.150Z)

```

1     # R1 Milestone ? Launch Report (windowing default flip: W1 cap=6 + R4)
2
3     **Status:** LAUNCHED 2026-06-18 (PR #204). **Flip:** the local-windowing defaults go ON
?
4     `max_window_duration 0?6`, `window_inactivity_end 0?1.5`, `inactivity_rally_thresh
0.08?0.20`,
5     `inactivity_min_density 0.34?0.30`. **W2 net-crossings, W1.5 hard-cap, and R5 blob-
fallback all stay OFF.**
6     This is the [Launch Report convention](../..) record made **before** the flip merged ?
total quality impact
7     on **both rulers** + the over-seg guard + honest caveats, for posterity.
8
9     > **Why this flip is safe to ship:** measured on the **n=13** golden corpus (the grown

```

```

set, incl. the two
10     > Boxhill clips GX010137/GX010141), the milestone clears the **net over-seg promotion
bar** with room to
11     > spare, and the verdict was **independently reproduced** (a different harness, #214)
and **adversarially
12     > stress-tested** (5-lens L00/seed/per-clip workflow). Nothing in the verification
refuted it.
13
14     ---
15
16     ## 1. Total quality impact ? BOTH rulers (n=13, leave-one-video-out unit = whole video)
17
18     The headline promotion metric is the **R2 task-faithful tolerance-match F1** (`tolF1`,
asymmetric ?_start=2/
19     ?_end=1.5); **tIoU-F1@0.5** is kept as the secondary continuity ruler.
20
21     | config | **R2 tolF1** | **tIoU-F1@0.5** | mean \?end\ | |
22     |---|---|---|---|
23     | baseline (all-OFF) | 0.137 | 0.236 | 44.6 s |
24     | + W1 cap=12 | 0.210 | 0.382 | 5.7 s |
25     | + cap=6 (R3) | 0.320 | 0.436 | 5.0 s |
26     | **+ R4 (milestone, shipped)** | **0.359** | **0.474** | **3.3 s** |
27
28     **Paired sign-tests (the promotion evidence):**
29
30     | transition | ? tolF1 (95% CI) | sign (W/L/T) | p | ? tIoU-F1 | sign | p |
31     |---|---|---|---|---|---|---|
32     | **baseline ? milestone** | **+0.222** [+0.152, +0.286] | **12/0/1** | **0.0005** |
+0.238 [+0.159, +0.318] | 12/0/1 | 0.0005 |
33     | cap12 ? cap6 (R3) | +0.110 [+0.070, +0.153] | 12/0/1 | 0.0005 | ? | ? | ? |
34     | cap6 ? +R4 | +0.040 [+0.015, +0.065] | 9/1/3 | 0.0215 | ? | ? | ? |
35
36     The single non-win in baseline?milestone is **mahadevpura_1** (the continuously-active
blob: tolF1 0.000?0.000,
37     a tie ? it needs R5, which stays OFF). **Both new clips improve:** GX010137 0.448?0.685,
GX010141 0.333?0.431.
38
39     ## 2. Over-segmentation guard (the promotion gate)
40
41     The R1 **net** over-seg guard (`tolerance_metrics.over_seg_guard`) is the gate (not the
strict per-video count
42     rule). Baseline ? milestone:
43
44     - **NET verdict: PASS** ? weighted (merge:split = 2:1) merge_rate/split_rate cost
**1.303 ? 0.429** (? ?0.874).
45     - **mean merge_rate 0.651 ? 0.161** ? drops on **all 13** videos; **zero per-video
merge_rate regressions**.
46     - **strict_passes = False BY DESIGN** (not a regression): the raw per-video *count* rule
trips because splitting
47     a mega-window lifts split_count from ~0 and the raw merge *count* is non-monotonic
(one window hiding 40
48     rallies = count 1/rate 1.0 ? bounded into pieces gluing 2 neighbours = count 9/rate
0.5 ? **fewer** rallies
49     hidden). The guard scores the recall-meaningful *rate*; see
[QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) "R1".
50
51     ## 3. Verification provenance (independent + adversarial)
52
53     - **Independent reproduction (#214 `lovo_resweep.py`, a different harness):** milestone
vs baseline 0.137?0.359,
54     sign **12-0-1, p<0.001**; R4 on-vs-off 0.320?0.359, sign **9-1-3, p=0.021** ? matches
to 3 decimals. Its LOVO
55     cap-selection picks **cap=6 unanimously (13/13)** on tIoU.
56     - **5-lens adversarial workflow** (reproduce / leave-one-out / R4-loss audit / new-clip
sanity / guard audit ?

```

57 synthesis): verdict ****STANDS****. Every L00 fold (n=12) clears CI-excludes-0 AND p<0.05;
worst fold (drop
58 GX010128, the biggest gainer) still ?+0.202, p=0.0010 ? ****no single clip drives the
win****.
59
60 ## 4. Honest caveats (recorded so they aren't relitigated)
61
62 1. ****The W1 cap is the dominant lever; R4 is a marginal increment.**** baseline?cap6 is
~+0.18 of the +0.222
63 tolF1 gain; the R4 inactivity-end trim adds a **small but significant** +0.040
(p=0.0215, sign 9/1/3). Frame R4
64 as "a small, significant end-precision gain" (mean |?end| 4.96?3.31 s), not a co-
equal driver.
65 2. ****R4's one loss is GX010137**** (tolF1 0.727?0.685, ?0.042) ? a tolerance-accounting
artifact, ****not**** an R4
66 failure: on that clip strict F1 **improves** (0.805?0.822), |?end| **improves**
(0.447?0.439), over-seg unchanged;
67 R4 simply trimmed 4 trailing predicted segments (43?39, toward gt=34). The clip's
overall milestone delta is
68 a large win (+0.237).
69 3. ****GX010141 is the worst-behaved clip post-milestone**** (merge_rate still 0.448, ~30 s
windows vs gt max ~10.5 s)
70 ? partially-merged, ****not**** a regression (it improves +0.098) and it does not break
the guard, but it's a
71 future over-merge target as the corpus grows.
72 4. ****cap=4-vs-6 wobble on tolF1**** (from #214): cap=4 edges cap=6 by ~0.005 (noise) ? but
****LOVO re-fitting to**
73 cap=4 generalizes **worse*** (??0.004) and tIoU is firmly cap=6, so ****cap=6 stands****.
Watch as n grows; switch
74 only if cap=4 pulls ahead **consistently across folds**.
75 5. ****n=13 is small.**** "Robust" here means "no single clip or bootstrap seed flips the
verdict," not large-sample
76 certainty. (Bootstrap CIs jitter ~0.001?0.004 run-to-run; means/sign-tests/p-values
are exact.)
77 6. ****Nightly re-baselined + an all-OFF floor added (DONE).**** The flip made the nightly's
`default` config
78 (= SERVED) equal `milestone`, so its config-drift detector flagged the frozen
baseline STALE ? re-snapshotting
79 in #216. A pinned ****baseline_off**** reference config was then added (follow-up) so
every nightly report keeps
80 showing the delta the milestone buys (floor tolF1 ****0.137**** vs milestone ****0.359****) ?
the launched win has to
81 keep standing its ground, not be taken at launch-time value.
82 7. ****Pure-logic eval only**** (CPU, persisted trajectories) ? no GPU/real-video re-
detection was run; the flip
83 changes only the windowing knobs the served path already reads.
84
85 ## 5. What did NOT flip (still default-OFF, retained)
86
87 - ****W2 net-crossings gate**** ? no ground-truth-backed lift (the weak-signals-retained
directive keeps it).
88 - ****W1.5 hard-cap**** ? over-segments normal clips (chops before R4); superseded by R5 for
the blob case.
89 - ****R5 blob-fallback**** ? a stratum tool for continuously-active clips; turn on
selectively, not globally.
90

15. assistant (2026-06-19T17:18:35.864Z)

Now let me run the cross-reference greps. I need to check "cited as live" for each filename in
CLAUDE.md, NEXT_STEPS.md, README.md and other top-level docs, plus count inbound references
repo-wide.

16. user (2026-06-19T17:18:37.442Z)

Found 33 files
docs\README.md
docs\R2_EVAL_METRIC_DESIGN.md
docs\QUALITY_ITERATIONS.md
docs\NEXT_STEPS.md
docs\CODE_MAP.md
docs\BACKLOG.md
CLAUDE.md
backend\providers\gemini.py
docs\DESKTOP_APP_PLAN.md
backend\main.py
backend\infrastructure\database.py
backend\api\job_worker.py
docs\archives\checkpoints\2026-06-14-hardening-loop-complete.md
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
docs\archives\HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
tests\test_async_jobs.py
tests\test_job_worker_loop.py
tests\test_validators_base.py
docs\PLATFORM_ARCHITECTURE.md
docs\CODE_AUDIT_AND_TEST_HARDENING.md
docs\COMMERCIALIZATION.md
docs\DEFERRED_HARDENING_PLAN.md
docs\EXECUTION_POLICY.md
docs\HARDENING_LOOP_HANDOFF.md
tests\test_job_self_scheduling.py
tests\test_execution_policy.py
backend\utils\run_telemetry.py
backend\infrastructure\execution_policy.py
docs\archives\decisions\DECISIONS.md
docs\UI_PROPOSAL.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md

17. user (2026-06-19T17:18:37.798Z)

```

1      # Next steps (live) ? owned-model + multisport
2
3      **Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness
design docs merged,
4      owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
5      ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
6      `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;
**PR-2
7      upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
8      compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
9      coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
10     `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
11     **Authoritative roadmap:**
12     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
13     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
14
15     ## ? Next-session / GPU-box pickup (2026-06-15)
16     **M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**),
train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16
drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already

```

present + suite-green** (`backend/eval/fusion\_compare.py` + `ablation.py`: person-driven ?F1 / bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level, not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing ? Actions minutes / spending limit).

17

18 **Immediate next step ? the real P3 LOVO measurement** (this box has a GPU + `torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden features + `.rallies.csv` labels are not yet here):

19 1. **[owner / GPU box]** `python -m backend.eval.fusion\_golden --manifest output/human\_lovo\_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6" ? produces `_golden_features.json` for the 6 golden videos. (GPU + WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)

20 2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
[`GOLDEN\_DATA\_SHARING.md`](GOLDEN_DATA_SHARING.md); tracked in **##175**.

21 3. **[CPU dev/agent session]** `python -m backend.eval.fusion\_compare --features-dir <shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir <shared>...\--granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to also pass the `skipif` litmus tests (`tests/test\_ablation.py::test\_litmus\_reproduces\_gate\_a`).

22

23 **Also queued (owner-gated / pending input):**

24 - Ship-gate target calibration ? **##172** (owner decision; needs corpus growth).

25 - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent will do it).

26 - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config bug.

27

28 **Leaving (bulky / M2 ? no start without owner go):** ActionFormer (M0.4), event-index DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack deprecation, multispot, repo rename, PMF execution.

29

30 ## Where we are (one paragraph)

31 ADR-008 ratified ("repo split driven by productionization, not isolation"): training lives in-repo (`training/`)

32 behind a CI-enforced import wall; the featurizer is single-sourced (`backend/features/rally_seq.py`, `train==serve`);

33 the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest output/human_lovo_manifest.json --floor`

34 `eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0 (LOVO **##0.626**, floor passed,

35 sign test 5/6 wins $p=0.109$? honestly not significant, `ship=False`). The **n=6** owned corpus is in the collector

36 (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet reclassification). The **Gemini**

37 battery is done** (see table) ? the moat is quantified.

38

39 ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)

40 | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |

41 |---|---|---|

42 | Heuristic (velocity windowing) | 0.657 | 0.157 |

43 | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |

44 | Two-stage WASB+Gemini refine | 0.83 | ? |

45 | **Gemini wrapper** (`gemini-flash-latest`) | **##0.93** | **##0.66** |

46

47 **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-court** drops the commodity

48 wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game pickups + dead-time merges,

49 the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-growth-sized gap, not an

50 architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG tapes (uploads candidate clips

51 only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry load-bearing.

52 Baselines tracked in `eval\_baselines/` (heuristic_lovo_n6.json,
gemini_wrapper_2026-06-10.json).

53

54 **## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation
in progress)**

55 ****This is the track a desktop/local agent is slated to pick up.**** Design docs landed (PR
#112 + session). All implementation is ****owner-gated**** (per explicit in the plan); each step =
ONE PR off `master`.

56

57 ****Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs
#156/#157/#160/#161/#163 + #165 + final records).**** See archived
`docs/archives/CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every
pre/post/review) and `docs/archives/HARDENING\_LOOP\_HANDOFF-COMPLETED-2026-06-14.md` (full rules
+ STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive
done; clean slate.

58

59 - ****Docs:**** `docs/MULTI\_SIGNAL\_FUSION\_PLAN.md` (shuttle + person + audio, one fusion
layer) .

60 `docs/EXPERIMENT\_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .

61 `docs/ROORA\_LABELING\_GUIDELINES.md` (v8) . `docs/HOW\_RALLY\_DETECTION\_WORKS.md`
(2-layer mental model).

62 - ****Key finding (no re-investigation needed):**** the owner's "held-shuttle counted as a
rally" bug is ****NOT a missing capability**** ? `backend/pipeline/detectors/rally\_gate.py` already
implements the net-crossing ****"Gate A" (apply_gate(..., min_crossings=2))** that kills
service-feed/held-shuttle windows, but it is ****only called from eval drivers**** (and there was no
`min\_crossings` knob in `IndexingConfig`). ****Wiring Gate A into the live segmenter (now via
fusion_hybrid) was the single cheapest, highest-confidence quality win.**** (P2 FusionSegmenter
specifics: registry `fusion\_hybrid`, ****default-OFF****, seeds from substrate, persists
`net\_crossings` + optional person features, optional served Gate A/B via config knobs;
adversarially reviewed; suite green. See `tests/test\_served\_gate\_a\_regression.py` for the honest
finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in
with knob for safety, and the leaderboard revert note.)

63 - ****Current status (2026-06-14):**** P1 (person cue extraction to reusable
`PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment
harness P1 DONE (adversarial review + NO-REGRESSION; suite green). ****P3 next (this PR + follow-
ups):**** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos;
eager-person-compute optimisation); then P4 audio via hit_detector.

64 - ****Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-
OFF, promote only on measured LOVO):****

65 1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served
path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served
neutral on production windowing per litmus ? kept opt-in with knob for safety.)

66 2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant
(players_in_court + two_sided + colocation etc. from `fusion\_features`) on the 6 golden; add the
eager compute optimisation note/landing. Pure + harness (testable here).

67 3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.

68 - ****Discipline (from EXPERIMENT_HARNESS.md):**** every new cue ****flag-gated, default
OFF****; promote to default ****only on measured LOVO lift**** through `run\_regression.py` (exit
0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.

69 - ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the
heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden
features + `rallies.csv` labels, which are not yet present in this checkout.

70 - ****External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are
exhausted; full details + citations in RALLY_QUALITY_RESEARCH.md
\$5.6.**** An owner-commissioned 31-reference review of the quality report ****validated* the
methodology and surfaced these future levers (research inputs, not active work, not new
claims; verify citations before any submission):**

71 1. ****"First-mile" paper framing (highest-value):**** stroke-classifiers (BST/DSTA) + the
big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all ****assume a pre-trimmed
rally clip already exists**** ? we solve the ignored prerequisite. Pitch the paper as ****the
missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage.****

72 2. ****Ground R2 in Action-Spotting / Precise-Event-Spotting****
(SoccerNet/TenniSet/T-DEED) ? turns R2 into a defensible publishable contribution.

73 3. ****Add domain baselines for any submission:**** MonoTrack + BST on ShuttleSet / Fine-
Badminton (we have only heuristic + Gemini today).

74 4. ****Prefer TemporalMaxer over ActionFormer**** for the M0.4 scorer swap (parameter-free
max-pool, ~2.8× fewer GMACs, small-N robust) ? see "M0.4 next increment" below.

75 5. ****Court polygons ? pose ? the rally-END lever:**** masking on-court players can
resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator
noise.

76 6. ****Name the eval?serve lesson "pipeline distribution skew"**** ? a citable paper
"lesson."

77

78 ****Discipline reminder:**** owner approval before starting any owner-gated item; one PR per
step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.

79

80 **## Prioritized next steps**

81 1. ****[owner, in progress] GROW THE GOLDEN CORPUS**** ? the single highest-leverage move;
every lever below feeds on it.

82 Priority order for new videos: ****(a) multi-court badminton**** (the target distribution
AND where the ship target

83 lives), ****(b) ?3 matches per new sport ? table tennis first**** (WASB transfers at F1
0.909; pickleball labels are

84 corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c)
capture variety per

85 ``VIDEO_RECORDING_GUIDELINES.md``; ****adult sessions only for the training pool****
(consent guardrail). Per video:

86 label ? ``curate import-local --normalize --license Proprietary --fps <true fps>`` ?
re-run the Gen-0 harness.

87 The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is
reachable this month.

88 2. ****[owner decision] Set the ship-gate target**** ? now informed: floor = heuristic 0.480
(necessary), ****multi-court**

89 reference = wrapper 0.66** (the number that makes the owned model worth serving),
clean-footage ceiling = 0.93

90 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70
on multi-court folds with

91 paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.

92 3. ****M0.4 next increment (\$0/local):**** swap the Gen-0 LogReg for a lightweight temporal
head (MIT, minutes on the

93 2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video
threshold calibration (the

94 LOVO-vs-oracle gap is visible per fold in the harness output). ****Prefer**
TemporalMaxer** (parameter-free

95 max-pool, ~2.8× fewer GMACs, small-N robust) over ActionFormer/ASFormer per the
2026-06-17 external review

96 (``RALLY_QUALITY_RESEARCH.md`` \$5.6) ? better fit for the cheap/local/fast mandate.

97 4. ****Multisport thread:**** ingest Roora pickleball/TT CSVs (``import-local --normalize``);
merged-prototype LOVO across

98 badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-
on-WASB serving stays gated

99 by C3 (provenance multiplies per sport).

100 5. ****PMF validation (cheaper than any engineering month):**** C13 academy interviews (+
the two added questions:

101 "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot
at the warmest 2-3 academies

102 (consent at capture; demo served via the Gemini path or human-polish ? the reel
pattern exists:

103 ``output/MahadevpuraSingles_highlights.mp4``). ****Field kit COMPLETE (2026-06-12, owner**
ratified the

104 physical-visit approach; reframed product-feedback-first per owner spec):**

105 ``docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md`` (the recruiting flyer ? owner fills ?/contact
placeholders and recruits) +

106 ``docs/FIELD_VISIT_PLAYBOOK.md`` (the hired associate's visit playbook: coach + student
script, demo-after-Q6

107 rule, ground rules, Phase-2 record-and-show-back stretch goal) +
``docs/FIELD_VISIT_ANSWER_SHEET.md`` (per-visit

108 capture form incl. demo-reaction + student blocks) + ``docs/PMF_FIELD_SIGNALS.md``
(G/A/R anchors + ****pre-registered**

109 decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first

```

visit).
110     6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
111         deletion in AI Studio); **rename repo off "badminton"** (owner action,
112         in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
113
114     ## Pointers / artifacts
115     - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` . Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) .
116     Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` . Quality history:
117     `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
118     - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
119     manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
120     - Corpus: collector DB (system-of-record, 6 matches/262 rallies) . eval manifest
`output/human_lovo_manifest.json`
121     (gitignored) . durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` .
122     labels: `...\Collect\Golden Labelled\`.
123     - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
124     (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
125     with a MISLEADING "credits depleted" 429; ramp gradually.
126     - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
127     PLATFORM slice = async job model (single-tenant).
128

```

18. assistant (2026-06-19T17:18:41.671Z)

NEXT_STEPS confirms the hardening loop (CODE_AUDIT, HARDENING_LOOP_HANDOFF) is COMPLETE and top-level files are pointer stubs by design. Now let me check the specific live-citation contexts for DEFERRED_HARDENING_PLAN and EXECUTION_POLICY, and check README.md + the inbound reference counts precisely.

19. user (2026-06-19T17:18:42.772Z)

```

CLAUDE.md:14: `docs/PLATFORM_ARCHITECTURE.md`, `docs/DEFERRED_HARDENING_PLAN.md`. Also
`docs/ROADMAP.md`
CLAUDE.md:32: (`DEFERRED_HARDENING_PLAN.md` #16, single-tenant); the **owned-model track**
starts at a
backend\api\job_worker.py:1: """Async job worker ? drain/wake loop (DEFERRED_HARDENING_PLAN #16,
EXECUTION_POLICY.md P3).
backend\api\job_worker.py:26: # Async job executor (DEFERRED_HARDENING_PLAN #16). max_workers=1
on purpose: a single
tests\test_async_jobs.py:1: """Async job model (DEFERRED_HARDENING_PLAN #16).
docs\DEFERRED_HARDENING_PLAN.md:9: Full archived snapshot (the original plan + per-item
design/risks/verification): [`archives/DEFERRED_HARDENING_PLAN-
COMPLETED-2026-06-14.md`](archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md).
docs\DEFERRED_HARDENING_PLAN.md:11:_(This top-level file is a pointer stub kept so existing
links to `docs/DEFERRED_HARDENING_PLAN.md` still resolve; the content now lives in the
archive.)_
docs\COMMERCIALIZATION.md:36:| C9 | **Packaging / distribution** | **P0** | ? | `pyproject.toml`
(PEP 621, hatchling) added; the diagnostic `setup.py` renamed to `scripts/doctor.py`; `indexer`
console entry point; CI installs `pip install -e .[dev]`. torch/torchvision stay an out-of-band
`--index-url` install (PEP 621 can't pin CUDA wheels). | Implemented per
`DEFERRED_HARDENING_PLAN.md` (#7). Remaining: publish `obsreport` so the `reporting` extra is
pip-installable. |
docs\COMMERCIALIZATION.md:37:| C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? |
Async job model shipped (`DEFERRED_HARDENING_PLAN.md` #16, PR #87); auth/tenancy still future. |
Multi-user/platform strategy in `PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy,
orchestration). |
docs\COMMERCIALIZATION.md:211: `DEFERRED_HARDENING_PLAN.md`; reflected PR #50/#51 (input-
quality reports, value-layer

```

docs\COMMERCIALIZATION.md:222:- `DEFERRED\_HARDENING\_PLAN.md` ? packaging (C9), async (C12) designs (pending approval).

docs\EXECUTION_POLICY.md:115:**Links:** the async-job substrate is `docs/DEFERRED\_HARDENING\_PLAN.md` S3 (#16); multi-worker future is docs\DESKTOP_APP_PLAN.md:63:- **Backend packaging:** the Python/FastAPI backend bundles via **PyInstaller** or a frozen venv as a local "sidecar." Packaging was already modernized ? **PEP 621 `pyproject.toml` + an `indexer` console entry point** shipped (`DEFERRED\_HARDENING\_PLAN` #7, DONE, PR #167) `[verified]` ? so a CLI MVP has a real head start.

docs\DESKTOP_APP_PLAN.md:157:[Omitted long matching line]

docs\BACKLOG.md:49:[Omitted long matching line]

backend\infrastructure\database.py:181: # v2: async job model (DEFERRED_HARDENING_PLAN #16). One row per submitted

backend\infrastructure\database.py:482: # --- Async job store (DEFERRED_HARDENING_PLAN #16)

docs\archives\decisions\DECISIONS.md:502:- The UI alpha greenlights DEFERRED_HARDENING_PLAN **#16** as PR-1; the owner still gives an

docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:5:**Related:** [CODE_MAP.md](../CODE_MAP.md), [CLAUDE.md](../CLAUDE.md), [EXPERIMENT_HARNESS.md](../EXPERIMENT_HARNESS.md), [ABLATION_LAYER.md](../ABLATION_LAYER.md), [DEFERRED_HARDENING_PLAN.md](../DEFERRED_HARDENING_PLAN.md), [NEXT_STEPS.md](../NEXT_STEPS.md), [QUALITY_ITERATIONS.md](../QUALITY_ITERATIONS.md)

docs\PLATFORM_ARCHITECTURE.md:49:> [`DEFERRED\_HARDENING\_PLAN.md`](DEFERRED_HARDENING_PLAN.md). Commercial constraints come

docs\PLATFORM_ARCHITECTURE.md:226:[`DEFERRED\_HARDENING\_PLAN.md`](DEFERRED_HARDENING_PLAN.md) #16** (async job model) ? build

docs\PLATFORM_ARCHITECTURE.md:242: [`DEFERRED\_HARDENING\_PLAN.md`](DEFERRED_HARDENING_PLAN.md) #16 slice, weighing:

docs\PLATFORM_ARCHITECTURE.md:580:- [`DEFERRED\_HARDENING\_PLAN.md`](DEFERRED_HARDENING_PLAN.md) ? #16 async job model (the P0 infra slice), #7 packaging.

docs\UI_PROPOSAL.md:4:**Date:** 2026-06-10 . **Inputs:** PMF_MARKET_RESEARCH, COMMERCIALIZATION, PLATFORM_ARCHITECTURE, DEFERRED_HARDENING_PLAN (#16), ROADMAP, VIDEO_RECORDING_GUIDELINES, reporting/spec.yaml, current API + static UI.

docs\UI_PROPOSAL.md:54:| B1 | **Async job model ? DEFERRED_HARDENING_PLAN #16** | The committed next PLATFORM slice; `POST /api/process` ? 202 + `job\_id`, `jobs` table (user_version 12?), ThreadPoolExecutor, `GET /api/jobs/{id}`. **Approving this proposal = greenlighting #16.** A 5760 min blocking HTTP call cannot sit behind a tunnel UI. |

docs\UI_PROPOSAL.md:97:1. **Greenlight** alpha scope above (= greenlights DEFERRED_HARDENING_PLAN #16)?

backend\main.py:64:# Async job worker ? drain/wake loop (DEFERRED_HARDENING_PLAN #16, EXECUTION_POLICY.md P3)

backend\main.py:129: the job worker thread (DEFERRED_HARDENING_PLAN #16). Returns the same response dict

backend\main.py:246: runs on the worker (DEFERRED_HARDENING_PLAN #16). The row is created 'queued'; the

docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:15:- `docs/DEFERRED\_HARDENING\_PLAN.md` (remaining: train/eval separation guardrails)

docs\archives\checkpoints\2026-06-strategy-foundation\README.md:18:- **DEFERRED_HARDENING_PLAN.md** ? packaging, train/eval guardrails, async job model (#16) ? *pending approval*.

docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md:35:5. Packaging modernization (`pyproject.toml` + clean entry points). ? **NOT shipped** (no `pyproject.toml` on master as of 2026-06-10 ? this entry was aspirational; corrected here per attribution-honesty). Still tracked as COMMERCIALIZATION C9 / DEFERRED_HARDENING_PLAN #7.

20. user (2026-06-19T17:18:43.306Z)

backend\api\job_worker.py:1: """Async job worker ? drain/wake loop (DEFERRED_HARDENING_PLAN #16, EXECUTION_POLICY.md P3).

backend\api\job_worker.py:42: """Raised inside the pipeline to SELF-SCHEDULE a job (EXECUTION_POLICY.md P3,

backend\main.py:64:# Async job worker ? drain/wake loop (DEFERRED_HARDENING_PLAN #16, EXECUTION_POLICY.md P3)

backend\main.py:247: worker drains it via `claim\_due\_job` (EXECUTION_POLICY.md P3 self-scheduling loop).

backend\main.py:634: # Resume self-scheduled work orphaned by the restart

```
(EXECUTION_POLICY.md P3): the
backend\utils\run_telemetry.py:11:(`docs/EXECUTION_POLICY.md`): the policy decides wait / abort
/ resume; this records the signals
backend\infrastructure\execution_policy.py:1:"""Per-job execution policy: bounded, observable
waits for long/flaky ops (EXECUTION_POLICY.md P1).
backend\infrastructure\execution_policy.py:45:      """Declarative resilience knobs, attachable
per job/op (JSON-serialisable; see EXECUTION_POLICY.md)."""
backend\infrastructure\database.py:208:          # v3: self-scheduling execution policy
(EXECUTION_POLICY.md P2). `policy` = the
backend\infrastructure\database.py:491:          (EXECUTION_POLICY.md) the worker honours; None ?
the worker's default policy."""
backend\infrastructure\database.py:550:      # --- Self-scheduling execution policy
(EXECUTION_POLICY.md P2) -----
backend\infrastructure\database.py:573:          schedule a wake-up (EXECUTION_POLICY.md P3 self-
scheduling). Returns 0.0 if one is
backend\providers\gemini.py:80:          # the worker forever). Per-poll logs at DEBUG (no INFO
spam); see EXECUTION_POLICY.md.
docs\CODE_MAP.md:114:[Omitted long matching line]
docs\CODE_MAP.md:216:[Omitted long matching line]
tests\test_execution_policy.py:1:"""High-coverage tests for
backend.infrastructure.execution_policy (EXECUTION_POLICY.md P1).
tests\test_job_self_scheduling.py:1:"""Tests for the v3 self-scheduling job substrate
(EXECUTION_POLICY.md P2).
tests\test_job_worker_loop.py:1:"""P3 worker self-scheduling loop (EXECUTION_POLICY.md P3).
```

21. assistant (2026-06-19T17:18:52.108Z)

This is decisive. Key findings:

- **DEFERRED_HARDENING_PLAN.md** ? the top-level is already a pointer stub, BUT it is cited as a LIVE/authoritative source in CLAUDE.md (line 14 "Read these first", line 32 "committed next PLATFORM slice = #16"), PLATFORM_ARCHITECTURE.md (multiple), COMMERCIALIZATION.md, UI_PROPOSAL.md, DESKTOP_APP_PLAN.md. This is exactly the contradiction trap. The stub itself is intentional and heavily inbound-linked ? it must stay so links resolve. KEEP_LIVE (the stub is the live pointer; do not touch).
- **EXECUTION_POLICY.md** ? Status "P1 SHIPPED; P2 + P3a landed; P3b next; P4 owner-gated." This is IN-PROGRESS with remaining slices. Heavily referenced by live code (backend/main.py, job_worker.py, database.py, gemini.py, 3 test files). KEEP_LIVE.

Now let me check the README.md index entries and the live-citation status for REAL_FOOTAGE_VALIDATION_REPORT and R1_MILESTONE_LAUNCH_REPORT.

22. user (2026-06-19T17:18:53.176Z)

```
docs\QUALITY_ITERATIONS.md:107:the **[R1 Milestone Launch
Report](R1_MILESTONE_LAUNCH_REPORT.md)** (per the Launch-Report convention).
docs\QUALITY_ITERATIONS.md:132:0-match outcome
[`REAL_FOOTAGE_VALIDATION_REPORT.md`](REAL_FOOTAGE_VALIDATION_REPORT.md) §5 R5 names).
docs\QUALITY_ITERATIONS.md:264:on two of them. Full report:
[`REAL_FOOTAGE_VALIDATION_REPORT.md`](REAL_FOOTAGE_VALIDATION_REPORT.md) (adversarially
docs\R2_EVAL_METRIC_DESIGN.md:7:the day-1 validation that motivates it:
[REAL_FOOTAGE_VALIDATION_REPORT.md](REAL_FOOTAGE_VALIDATION_REPORT.md);
```

23. user (2026-06-19T17:18:54.903Z)

29:[Omitted long matching line]

24. assistant (2026-06-19T17:18:59.053Z)

Let me see the README.md index line and check whether REAL_FOOTAGE / R1_MILESTONE roadmap items (R2?R8) are still being tracked as open work, which determines whether REAL_FOOTAGE is terminal.

25. user (2026-06-19T17:19:00.012Z)

```

20     - Evaluation harness and calibration drivers (in `backend/eval/`)
21     - [GOLDEN_DATA_SHARING.md](GOLDEN_DATA_SHARING.md) ? How the GPU box and CPU dev/agent
sessions share golden artifacts (trajectory CSVs, `_golden_features.json`) via a OneDrive
folder; the GPU-extract ? CPU-litmus split. See **#175**.
22
23     ## Conventions & templates
24     - [PROJECT_DOC_TEMPLATE.md](PROJECT_DOC_TEMPLATE.md) ? Reusable skeleton for a **tracked
project doc25: status header · decisions-locked · progress tracker (??/?) · definition-of-done
· archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7
and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.
25
26     ## Historical documents
27     Completed plans, old research, ADRs, and checkpoints live in [`archives/`](archives/).
28
29     - [CODE_AUDIT_AND_TEST_HARDENING.md](CODE_AUDIT_AND_TEST_HARDENING.md) ? **COMPLETED**
(2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing
/ contract / architectural audit, 626-test infrastructure, positive test plans for the 5
identified issues, multi-layer regression protocol (including ablation/experiment layer), and
pre/post-change verification checklist. See `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-
COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md`
for the final archived state. (Top-level is now pointer stub per review; full content in
archives/.)
30
31     See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md`
for classification of small tech debt / starter projects.

```

26. user (2026-06-19T17:19:00.827Z)

```

100     truths once instrumented: single-court count ?, multicourt ? (needs a court signal),
strict-precision ?
101     (needs merge control). Default-OFF lens + knob; the owner decides any flip on the
numbers. *(P7 in
102     [`RALLY_DETECTION_GAP_CLOSURE.md`](RALLY_DETECTION_GAP_CLOSURE.md) §7.)*
103
104     ### R1 ? MILESTONE LAUNCHED (cap=6 + R4 flipped default-ON) ? re-verified at n=13
*(2026-06-18)*
105     The R1 milestone is **shipped** (PR #204): the W1 cap=6 + R4 inactivity-end defaults are
now ON.
106     Full quality-impact table on both rulers, the over-seg guard, and the honest caveats are
recorded in
107     the **[R1 Milestone Launch Report](R1_MILESTONE_LAUNCH_REPORT.md)** (per the Launch-
Report convention).
108
109     **Re-verified on the grown n=13 corpus** (the 2 new Boxhill clips GX010137/GX010141) ?
the milestone
110     holds and the headline is *stronger* than at n=11:
111
112     | transition | n=11 (filed) | **n=13 (launched)** |
113     |---|---|---|
114     | baseline ? milestone tolF1 | ?+0.232, 10/0, p=0.002 | **?+0.222, 12/0/1, p=0.0005** |
115     | cap12 ? cap6 tolF1 | ?+0.108, 10/0, p=0.002 | **?+0.110, 12/0/1, p=0.0005** |
116     | cap6 ? +R4 tolF1 | ?+0.047, 8/0, p=0.008 | ?+0.040, **9/1/3**, p=0.0215 |
117     | net over-seg guard | PASS (mr 0.694?0.150) | **PASS** (mr 0.651?0.161, 0 regressions)
|
118
119     Both new clips improve (GX010137 0.448?0.685, GX010141 0.333?0.431); tIoU ladder
baseline 0.236 ?
120     milestone 0.474. Independently reproduced by a 2nd harness (#214 `lovo_resweep.py`,
matches to 3 dp;
121     LOVO picks cap=6 **13/13** on tIoU) and survived a 5-lens adversarial workflow (every
LOO fold clears
122     the bar; worst-fold drop GX010128 still ?+0.202, p=0.0010 ? no single clip drives it).
**Honest notes:**
123     W1 cap is the dominant lever / R4 is the marginal +0.040 increment (1 loss = GX010137, a
tolerance

```

```

124     artifact); the gate is the NET rate-based guard (strict=False by design); GX010141 is
the worst-behaved
125     clip (future over-merge target). The nightly self-flags STALE post-flip ? re-baseline
once with
126     `--update-baseline`. W2 / hard-cap / R5 stay default-OFF.
127
128     ### R5 ? blob-fallback for the continuously-active blob (default-OFF, stratum tool)
*(2026-06-17)*
129     The one clip the cap6+R4 milestone still fails is the **continuously-active blob**
(mahadevpura_1:
130     ~174 s of play with no inactivity gap to split on). The W1 gap-split finds no gap; R4
finds no slow
131     tail; the result is one ~66 s mega-window that matches nothing (**tolF1 0.000**, the
degenerate
132     0-match outcome [ `REAL_FOOTAGE_VALIDATION_REPORT.md` ](REAL_FOOTAGE_VALIDATION_REPORT.md)
$5 R5 names).
133
134     **R5** ( `window_blob_fallback`, default-OFF) is the last resort: **after** the gap-split
and the R4
135     trim have each had their chance, any group still longer than `window_blob_factor` × the
cap is a
136     genuine blob ? midpoint-chop it to ? the cap, **log the forced cut**, and **flag every
resulting
137     window `forced_cut=True`** (surfacing the clip as one that needs rally-STATE cues ? net-
crossings /
138     two-sided motion ? not a bigger cap). `_blob_fallback_chop` in `tracknet_runner`, wired
through
139     WindowingPreset / IndexingConfig / the served path / `rally_seg_eval --blob-fallback`.

```

27. assistant (2026-06-19T17:19:17.358Z)

I now have enough to judge all six. Let me assess the two remaining reports against terminal-state criteria.

****REAL_FOOTAGE_VALIDATION_REPORT.md**** ? Status: "validation complete, roadmap proposed (all flags default-OFF)." It is a dated (2026-06-16) validation report. But critically, its \$5 roadmap (R1?R8) is being actively executed: R1 shipped (PR #204), R2 has its own design doc (R2_EVAL_METRIC_DESIGN.md), R4 shipped with R1, R5 is the named lever for the blob case. QUALITY_ITERATIONS.md \$R5 cites it as the source naming R5, and R2_EVAL_METRIC_DESIGN.md cites it as "the day-1 validation that motivates it." The validation itself is complete, but the doc is the live root-cause/roadmap backbone that downstream active work (R2, R5, R6) still references and folds from. This is ambiguous ? the validation portion is terminal but the roadmap is live and cited as the motivating/naming source. Verdict: ASK_OWNER.

****R1_MILESTONE_LAUNCH_REPORT.md**** ? Status: "LAUNCHED 2026-06-18 (PR #204)." A Launch Report is by convention a posterity record made before a flip merges. The flip has merged. This is a terminal launch record (the convention explicitly is "recorded for posterity"). It is cited as a live source by QUALITY_ITERATIONS.md \$R1 ? but as a posterity pointer to a completed launch, not as a next/in-progress plan. By the Launch-Report convention these are archival posterity records (top-level "archives/" destination for one-off launch records). However it has no already-archived copy yet, was launched only yesterday (2026-06-18), and remains an actively-cited record of the current default config. Given it documents the CURRENT live default config and is freshly cited, archiving it one day post-launch risks churn ? but its lifecycle IS "launched/terminal." This is the cleanest ARCHIVE_NOW candidate among reports, though I'll flag it as a launch record that the convention says lives in archives/ top-level. Given it is freshly cited as the live record of current defaults in QUALITY_ITERATIONS, I'll mark ARCHIVE_NOW with the destination as a launch record but note the recency. Actually, the safer read: it documents the live config and was launched 1 day ago; the launch-report convention records it "for posterity" but does not say archive immediately. Since it is cited live and very fresh, I'll set ASK_OWNER would be over-cautious ? the convention is clear these are posterity records. I'll go ARCHIVE_NOW to archives/ top-level (launch records).

Let me get exact inbound reference counts for the two reports to finalize.

28. user (2026-06-19T17:19:17.923Z)

docs\QUALITY_ITERATIONS.md:1

Found 1 total occurrence across 1 file.

29. user (2026-06-19T17:19:18.513Z)

docs\R2_EVAL_METRIC_DESIGN.md:1

docs\QUALITY_ITERATIONS.md:2

Found 3 total occurrences across 2 files.

30. assistant (2026-06-19T17:19:29.129Z)

I have all the evidence needed. Let me compile the structured verdicts.

Summary of my analysis:

1. **CODE_AUDIT_AND_TEST_HARDENING.md** ? Already a thin pointer stub (per PR #165), points to its archived COMPLETED copy. Intentional stub ? KEEP_LIVE. README.md "Historical documents" links to it.
2. **DEFERRED_HARDENING_PLAN.md** ? Already a pointer stub, but CONTRADICTION TRAP: cited as LIVE/authoritative in CLAUDE.md ("Read these first" + "committed next PLATFORM slice = #16"), PLATFORM_ARCHITECTURE.md, COMMERCIALIZATION.md, UI_PROPOSAL.md, DESKTOP_APP_PLAN.md. Heavy inbound link load ? the stub must stay so links resolve. KEEP_LIVE (it's an intentional stub serving live citations).
3. **HARDENING_LOOP_HANDOFF.md** ? Already a thin pointer stub (per PR #165). Intentional. KEEP_LIVE.
4. **EXECUTION_POLICY.md** ? IN-PROGRESS (P1 shipped, P2/P3a landed, P3b is NEXT, P4 owner-gated). Heavily referenced by live production code + tests. Not terminal. KEEP_LIVE.
5. **REAL_FOOTAGE_VALIDATION_REPORT.md** ? Validation complete, but its R1?R8 roadmap is actively being executed (R1/R4 shipped, R2 has its own design doc, R5 named here, R6 pending) and it's cited as the live motivating/naming backbone by QUALITY_ITERATIONS.md and R2_EVAL_METRIC_DESIGN.md. Ambiguous (terminal validation vs live roadmap). ASK_OWNER.
6. **R1_MILESTONE_LAUNCH_REPORT.md** ? LAUNCHED 2026-06-18 (terminal launch record per the Launch-Report convention = posterity record). One inbound ref (QUALITY_ITERATIONS \$R1, as a posterity pointer). Cleanest terminal candidate, but launched only 1 day ago and documents the current live default config ? flagging the recency.

31. user (2026-06-19T17:20:05.436Z)

Structured output provided successfully

32. assistant (2026-06-19T17:20:17.010Z)

I have completed the audit of all six docs in my batch. Summary of verdicts:

- **CODE_AUDIT_AND_TEST_HARDENING.md** ? KEEP_LIVE. Already an intentional pointer stub (PR #165) to its archived COMPLETED copy.
- **DEFERRED_HARDENING_PLAN.md** ? KEEP_LIVE. Already a pointer stub, but it's the contradiction trap: CLAUDE.md cites #16 as the committed next platform slice, plus heavy inbound links (21 refs across CLAUDE.md, PLATFORM_ARCHITECTURE, COMMERCIALIZATION, UI_PROPOSAL, live code). The stub must stay so live citations resolve.
- **HARDENING_LOOP_HANDOFF.md** ? KEEP_LIVE. Already an intentional pointer stub (PR #165).
- **EXECUTION_POLICY.md** ? KEEP_LIVE. In-progress (P1 shipped, P2/P3a landed, P3b is the named NEXT slice, P4 owner-gated); wired into live production code + tests. No archived copy.
- **REAL_FOOTAGE_VALIDATION_REPORT.md** ? ASK_OWNER. Header says "validation complete" but it's the live root-cause/roadmap backbone (R1?R8); R2_EVAL_METRIC_DESIGN cites it as the motivating source and QUALITY_ITERATIONS \$R5 says it names R5. Validation event is terminal; the roadmap it carries is still live.

- ****R1_MILESTONE_LAUNCH_REPORT.md**** ? ASK_OWNER. Lifecycle is terminal (LAUNCHED 2026-06-18, PR #204) and the Launch-Report convention treats it as a posterity archive record ? but it launched only 1 day before this audit and is the live record of the current default config (cited in QUALITY_ITERATIONS SR1), so I flagged the recency rather than archiving unilaterally.

No doc in this batch is a clean unilateral ARCHIVE_NOW ? four are intentional stubs or live in-progress work (KEEP_LIVE), and the two reports each carry a contradiction (live citations / one-day-old launch record) that warrants owner confirmation.